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3. **Title of the invention:**

Smart Two-Wheeler Lock System for Anti-Theft Using Fingerprint Authentication

4. Technical field of the invention:

- Vehicle security systems.
- Biometric authentication technology.
- Embedded systems and microcontroller-based control.
- Smart locking and anti-theft systems for two-wheelers.

5. Prior art:

- Conventional two-wheelers primarily use **mechanical key-based locking systems** for ignition and vehicle access. These systems are vulnerable to key loss, duplication, lock picking, and unauthorized vehicle operation, thereby increasing the risk of vehicle theft.
- Existing vehicle security systems have proposed **biometric authentication** using fingerprint or iris recognition to enable vehicle access. For example, **US Patent No. US5686765A, "Vehicle Security System Including Fingerprint and Eyeball Part Identification,"** discloses a system that authenticates the driver using fingerprint or retinal recognition before enabling the ignition. However, the invention is primarily intended for automobiles and does not specifically address the security requirements of two-wheelers.
- **US Patent No. US6828899B2, "Fingerprint Identification System and Method for a Motor Vehicle,"** describes a fingerprint-based authentication mechanism that controls vehicle ignition, steering lock, and key functions. Although the invention improves access control, it does not integrate real-time theft notification, location tracking, or a dedicated anti-theft mechanism for motorcycles.
- **Chinese Patent CN1613696A, "Fingerprint Safety Device of Vehicle,"** discloses a fingerprint-based anti-theft system in which the vehicle door and engine can be operated only after successful fingerprint verification. While the invention enhances security through biometric authentication, it does not provide remote alerts, live GPS tracking, or a complete anti-theft solution tailored to two-wheelers.
- Although the above prior art improves vehicle security by incorporating biometric authentication, the disclosed systems generally focus on passenger automobiles or individual security functions. They do not provide a **cost-effective, integrated anti-theft solution for two-wheelers** combining fingerprint authentication, ignition control, alarm generation, and optional GSM/GPS-based owner notification and vehicle tracking.

6. Object:

- To provide a **smart and secure locking system** for two-wheelers using biometric fingerprint authentication.
- To eliminate the dependence on **traditional mechanical keys**, thereby reducing the risk of key loss or duplication.
- To allow access to the vehicle **only to authorised users** through fingerprint verification.
- To enhance the **security of two-wheelers** and minimise the chances of vehicle theft.
- To provide a **fast, reliable, and user-friendly** keyless locking and unlocking mechanism.
- To integrate a **microcontroller-based electronic control system** for efficient operation of the locking mechanism.
- To provide an option for **theft alerts and real-time notifications** through GSM technology in case of unauthorised access attempts.
- To support **GPS-based vehicle tracking** for locating the vehicle in case of theft.
- To develop a **cost-effective, compact, and energy-efficient** security system suitable for motorcycles and scooters.

- To provide a system that can be **easily installed and upgraded** with additional smart security features such as IoT connectivity or mobile application support.

7. Synopsis:

The present invention relates to a **Fingerprint-Based Smart Two-Wheeler Lock and Theft Prevention System** designed to improve the security of motorcycles and scooters through biometric authentication. The system replaces the conventional key-based locking mechanism with a fingerprint recognition system, ensuring that only authorised users can access and operate the vehicle.

The invention consists of a fingerprint sensor, a microcontroller (such as Arduino Uno), and an electronically operated locking mechanism. When a user places a registered fingerprint on the sensor, the system verifies the fingerprint against stored templates. Upon successful authentication, the microcontroller activates the locking mechanism, allowing the vehicle to be unlocked. If the fingerprint is not recognised, the vehicle remains locked, and optional security features such as a buzzer, GSM alert, or GPS tracking can be activated.

The proposed system enhances vehicle security by preventing unauthorised access, eliminating the need for physical keys, and reducing the risk of theft. It is cost-effective, easy to install, user-friendly, and suitable for motorcycles, scooters, and electric two-wheelers. The invention can also be expanded with IoT-based remote monitoring and control features, making it a practical and modern solution for vehicle protection.

8. Brief description of drawings (if any):

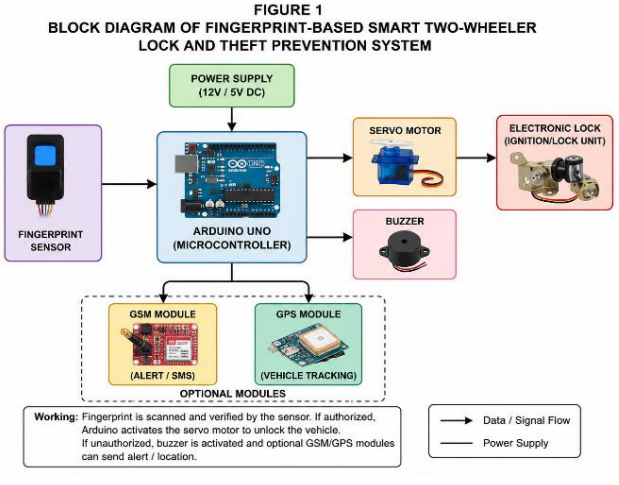


Figure 1

Figure 1 illustrates the block diagram of the proposed **Fingerprint-Based Smart Two-Wheeler Lock and Theft Prevention System**. It shows the connection between the power supply, fingerprint sensor, Arduino Uno, servo motor, electronic lock, buzzer, and optional GSM/GPS modules. The fingerprint sensor authenticates the user, the Arduino processes the data, and the servo motor unlocks the vehicle upon successful verification. In case of unauthorised access, the buzzer is activated, while the optional GSM and GPS modules provide alerts and vehicle tracking.

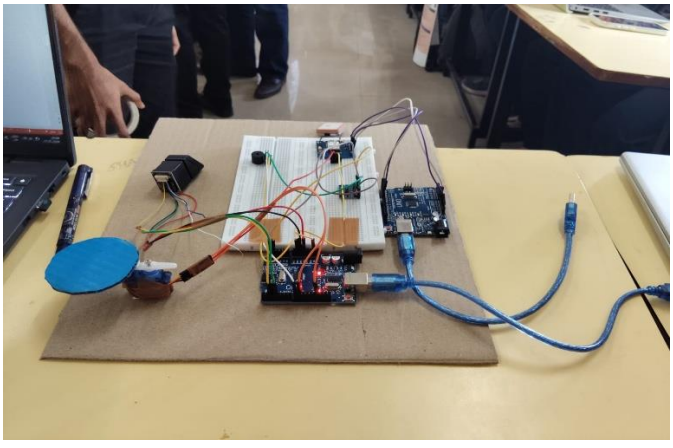


Figure 2

Figure 2. Demonstration of the proposed Smart Two-Wheeler Lock System for Anti-Theft prototype showing the Arduino Uno, fingerprint sensor, servo motor, relay module, and interfacing circuitry.

9. Detail description of the invention:

The present invention relates to a **Fingerprint-Based Smart Two-Wheeler Lock and Theft Prevention System** that enhances the security of motorcycles and scooters by replacing the conventional key-based locking mechanism with a biometric authentication system.

The system comprises a **fingerprint sensor**, **Arduino Uno microcontroller**, **servo motor**, **electronic locking mechanism**, **power supply**, **buzzer**, and optional **GSM** and **GPS** modules. The fingerprint sensor captures the fingerprint of the user and transmits the biometric data to the Arduino Uno for authentication. The microcontroller compares the scanned fingerprint with the pre-enrolled fingerprint templates stored in the system.

When a valid fingerprint is detected, the Arduino Uno sends a control signal to the servo motor. The servo motor actuates the electronic locking mechanism, allowing the vehicle to be unlocked and operated by the authorised user. Once the operation is completed, the lock can automatically return to its original locked position to ensure continuous security.

If the scanned fingerprint does not match any authorised template, the system denies access by keeping the lock engaged. Simultaneously, the buzzer is activated to indicate an unauthorised access attempt. If equipped with a GSM module, the system sends an alert message to the registered owner's mobile phone. The optional GPS module enables real-time tracking of the vehicle, allowing the owner to monitor its location in case of theft or unauthorised movement.

The entire system is powered by a regulated power supply, ensuring stable and reliable operation of all electronic components. The modular design allows easy installation on existing motorcycles and scooters and can be upgraded with additional features such as IoT connectivity, mobile application control, cloud-based monitoring, or remote diagnostics.

The proposed invention provides a **secure, reliable, cost-effective, and user-friendly keyless locking system** that significantly reduces the risk of vehicle theft while improving convenience for authorised users. It is suitable for motorcycles, scooters, electric two-wheelers, and other similar vehicles requiring enhanced security.

10. Best method of performance of the invention:

The invention is preferably performed by installing a **fingerprint sensor**, **Arduino Uno microcontroller**, **servo motor**, **electronic locking mechanism**, **power supply**, and an optional **buzzer**, **GSM**, and **GPS** module on a two-wheeler. Initially, the fingerprints of authorised users are enrolled and stored in the fingerprint sensor's memory. When a user places a finger on the sensor, the fingerprint is scanned and compared with the stored templates. If a match is found, the Arduino Uno processes the authentication result and sends a signal to the servo motor, which unlocks the electronic lock, allowing the user to access and start the vehicle.

For example, when the registered owner places their finger on the sensor, the system verifies the fingerprint within a few seconds and unlocks the vehicle automatically. If an unauthorised person attempts to access the vehicle using an unregistered fingerprint, the authentication fails, the lock remains engaged, and the buzzer is activated. If optional GSM and GPS modules are integrated, an alert message is sent to the owner's mobile phone and the vehicle's location can be tracked. This method provides a secure, reliable, and convenient keyless locking solution while significantly reducing the risk of vehicle theft.

11. CLAIMS:

- A **fingerprint-based smart locking system** for two-wheelers that allows access only to authorised users through biometric authentication.
- The system comprises a **fingerprint sensor**, **microcontroller (Arduino Uno)**, **servo motor**, **electronic locking mechanism**, and **power supply**.
- The fingerprint sensor captures and verifies the user's fingerprint against the stored authorised fingerprint templates.
- Upon successful authentication, the microcontroller activates the servo motor to unlock the electronic locking mechanism.
- If the fingerprint authentication fails, the locking mechanism remains engaged, preventing unauthorised access.
- The system includes a **buzzer** that generates an audible alarm when an unauthorised access attempt is detected.
- The system may include a **GSM module** to send alert messages to the registered owner's mobile phone in case of unauthorised access.
- The system may include a **GPS module** to provide real-time vehicle location tracking during theft or emergency situations.
- The system supports **multiple authorised users** by storing multiple fingerprint templates.
- The invention provides a **keyless, secure, reliable, cost-effective, and user-friendly** anti-theft solution suitable for motorcycles, scooters, and electric two-wheelers.

12. Inventive step of your invention:

- The invention replaces the conventional **mechanical key-based locking system** with **biometric fingerprint authentication**, allowing access only to authorised users.

- It integrates a **fingerprint sensor, microcontroller, and electronic locking mechanism** into a compact and cost-effective security system specifically designed for two-wheelers.
- Unlike traditional lock systems that can be bypassed by key duplication or lock tampering, the proposed invention uses **unique biometric fingerprints**, making unauthorised access significantly more difficult.
- The invention provides an **automatic electronic locking and unlocking mechanism**, improving user convenience while enhancing vehicle security.
- The system can be integrated with **GSM and GPS modules** to send theft alerts and provide real-time vehicle tracking, offering an additional layer of protection not commonly available in basic two-wheeler locks.
- The modular design allows the system to be **easily installed on existing motorcycles and scooters** without major modifications, reducing implementation cost.
- The invention is designed using **readily available, low-cost electronic components**, making it economical while maintaining reliable performance.

13. Industrial application:

- **Automobile Industry** – For enhancing the security of motorcycles, scooters, and electric two-wheelers.
- **Vehicle Security Systems** – As a biometric anti-theft solution for personal and commercial vehicles.
- **Electric Vehicle (EV) Industry** – For secure access control in electric bikes and scooters.
- **Fleet Management Services** – To provide secure access to authorised riders in delivery and rental fleets.
- **Two-Wheeler Rental Services** – For keyless and controlled vehicle access.
- **Embedded Systems and IoT Applications** – For integration with smart vehicle monitoring and tracking systems.
- **Public and Private Transportation** – To improve vehicle security and reduce theft.
- **Smart Mobility Solutions** – As a component of intelligent transportation and connected vehicle systems.

14. Abstract:

The present invention relates to a **Fingerprint-Based Smart Two-Wheeler Lock and Theft Prevention System** designed to enhance the security of motorcycles, scooters, and electric two-wheelers. The system employs a **fingerprint sensor**, **Arduino Uno microcontroller**, **servo motor**, and an **electronic locking mechanism** to provide secure, keyless access to authorised users. Upon successful fingerprint authentication, the microcontroller actuates the locking mechanism to unlock the vehicle, while unauthorised access attempts keep the vehicle locked and may trigger a buzzer. The system can also be integrated with **GSM and GPS modules** to send theft alerts and enable real-time vehicle tracking. The invention offers a **cost-effective, reliable, and user-friendly** solution that improves vehicle security, eliminates the need for conventional keys, and significantly reduces the risk of two-wheeler theft.

15. Drawing

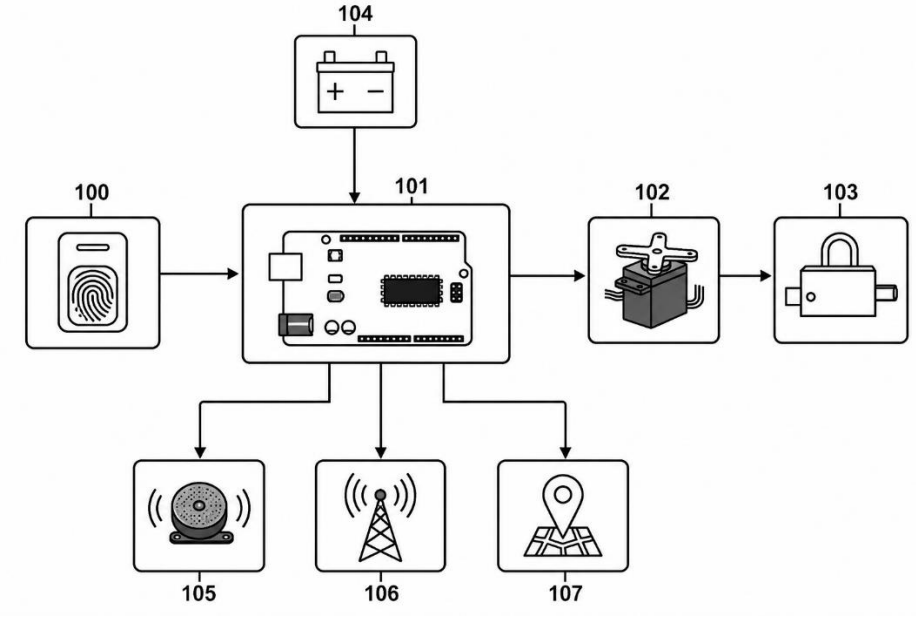


Figure 3

Figure 3 illustrates the block diagram of the proposed **Fingerprint-Based Smart Two-Wheeler Lock and Theft Prevention System**. The system comprises a **fingerprint sensor (100)**, a **microcontroller (101)**, a **servo motor (102)**, an **electronic locking mechanism (103)**, a **power supply (104)**, a **buzzer (105)**, and optional **GSM (106)** and **GPS (107)** modules. The fingerprint sensor (100) captures the user's fingerprint and transmits the biometric data to the microcontroller (101) for authentication. Upon successful verification, the microcontroller sends a control signal to the servo motor (102), which actuates the electronic locking mechanism (103) to unlock the vehicle. In the event of an unauthorised access attempt, the locking mechanism remains engaged, and the buzzer (105) is activated to generate an audible alarm. The optional GSM module (106) sends an alert message to the registered owner's mobile device, while the optional GPS module (107) provides real-time vehicle location tracking. The power supply (104) ensures reliable operation of all the system components, thereby providing a secure, keyless, and efficient anti-theft solution for two-wheelers.